

RPC muography comparison - plots for approval

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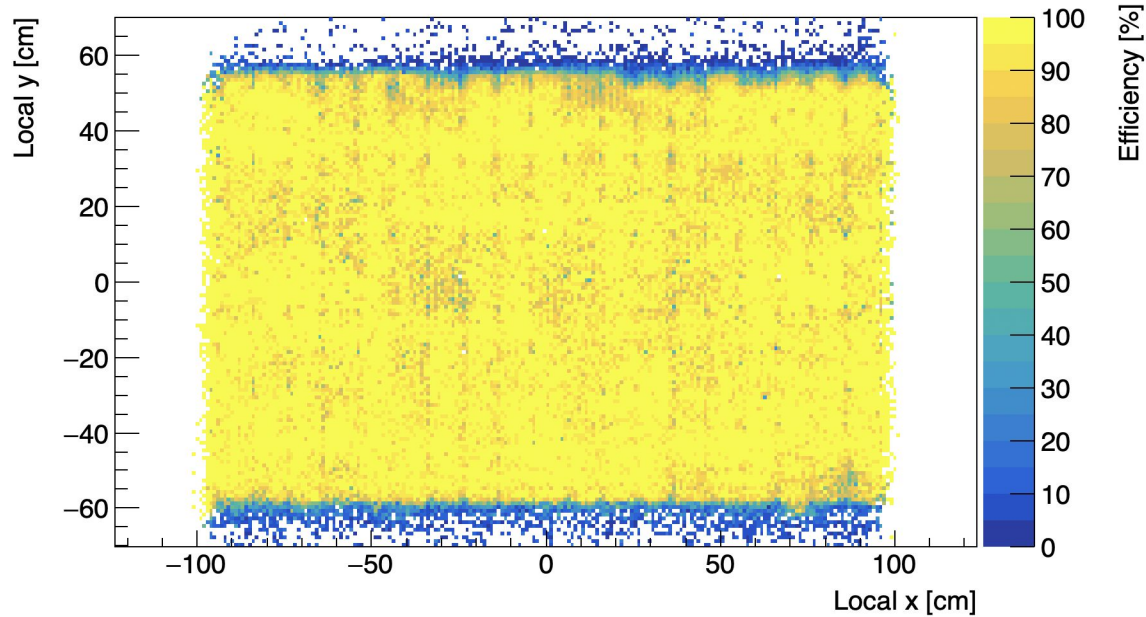
RPC DPG meeting, 01.07.2026

Introduction

- The RPC efficiency maps, co called muographies, have been obtained, exploring the Segment extrapolation method.
- The granularity of the muography plots is about $1 \times 1 \text{ cm}^2$.
- Two sets of proton-proton collision data from 2018 (Run-2) and 2025 (Run-3), taken at similar conditions, i.e. inst. luminosity $\sim 2.5 \text{E}34 \text{ cm}^{-2}\text{s}^{-1}$ and number of colliding bunches > 2000 have been used for the analysis.
- The muographies have been obtained for all the RPC rolls and the efficiency maps have been compared in order to study for any sign of potential ageing.
- A special attention is given to the RPC chambers, that have been switched Off because of the leaks, and later on have been repaired and switched again On.
- Apart from the visual comparison , a simple statistical analysis comparing 2018 and 2025 data has been performed as well.
- For every 2D bin ($\sim 1 \times 1 \text{ cm}^2$) a relative efficiency difference $\Delta \epsilon_{ij}$ has been calculated, exploring the following formula:

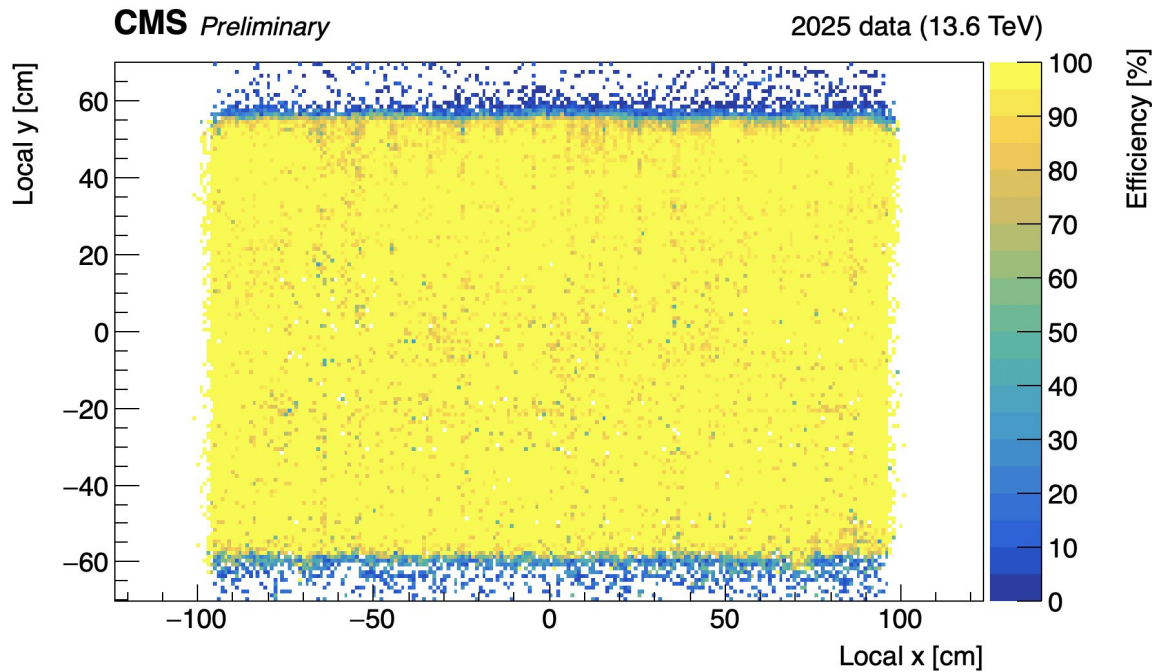
$$\Delta \epsilon_{ij} = \{\epsilon_{ij}(2018) - \epsilon_{ij}(2025)\} / \{\epsilon_{ij}(2018) + \epsilon_{ij}(2025)\}$$

- The distribution of $\Delta \epsilon_{ij}$ has been fitted to a Gaussian distribution

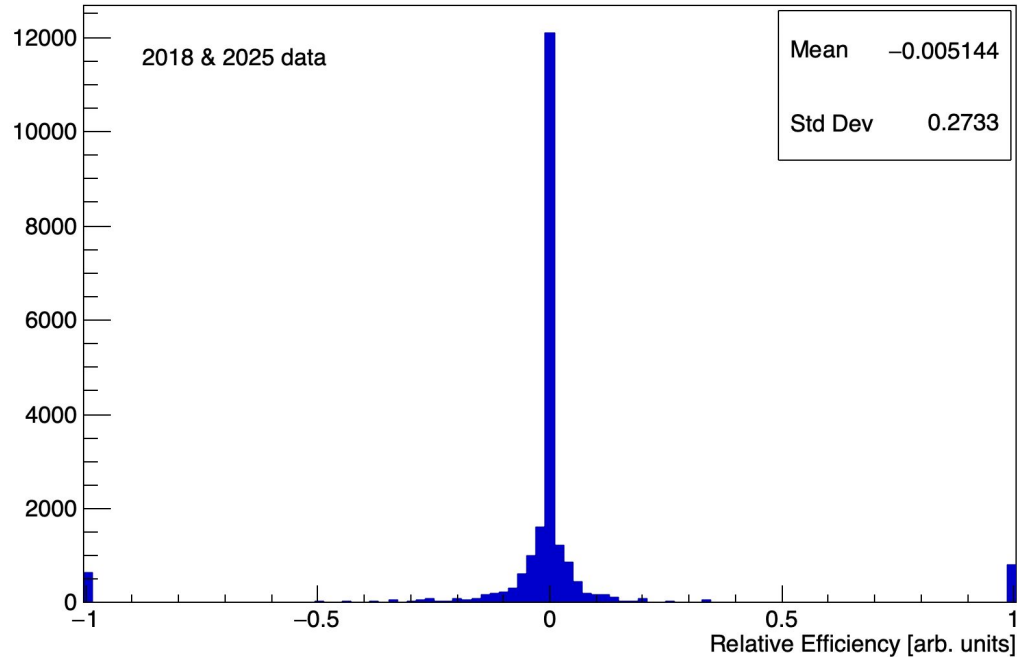


The plot represents the efficiency vs the local impact point on the RPC surface of a roll in one of the Barrel wheels (W-2_RB1in_S09_Backward) measured in 2018 (Run-2). The lower efficiency spots are due to the dead regions induced by spacers (the two resistive electrodes are separated by 2 mm spacers to form the parallel plate gas gaps). on grid with dimensions with $10 \times 10 \text{ cm}^2$. The Y axes is along the strip length. Due to the geometrical reasons there are not extrapolated hits in the area corresponding to $x > 100 \text{ cm}$ and because of this the efficiency is not calculated for it.

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The plot represents the efficiency vs the local impact point on the RPC surface of a roll in one of the Barrel wheels (W-2_RB1in_S09_Backward) measured in 2018 (Run-3) after the chamber has been connected again. The lower efficiency spots are due to the dead regions induced by spacers (the two resistive electrodes are separated by 2 mm spacers to form the parallel plate gas gaps) on grid with dimensions with $10 \times 10 \text{ cm}^2$. The Y axes is along the strip length. Due to the geometrical reasons there are not extrapolated hits in the area corresponding to $x > 100 \text{ cm}$ and because of this the efficiency is not calculated for it. Contact: [cms-dpg-conveners-rpc@cern](mailto:cms-dpg-conveners-rpc@cern.ch)[NOSPAMPLEASE.ch](mailto:NOSPAMPLEASE@cern.ch)

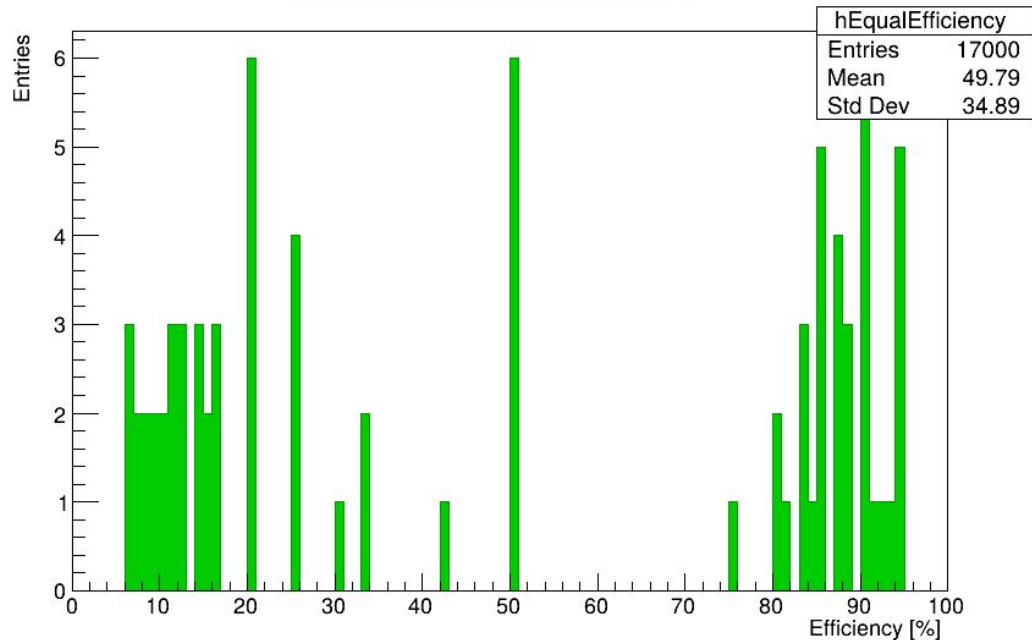


Relative efficiency difference distribution. A bin-by-bin comparison of the RPC efficiencies measured in 2018 and 2025 is performed using the normalized difference $(\text{Eff}_{2018} - \text{Eff}_{2025}) / (\text{Eff}_{2018} + \text{Eff}_{2025})$. This quantity is bounded between -1 and $+1$, where a value of 0 indicates identical efficiencies in the two data-taking periods. A value of $+1$ corresponds to bins where the 2025 efficiency is zero while the 2018 efficiency is non-zero, whereas -1 corresponds to bins where the 2018 efficiency is zero while the 2025 efficiency is non-zero. As a conclusion, the current results demonstrate that the chambers have shown a good level of efficiency after being put back into operation following the repair.

Spare slides

Complimentary slides

NOT FOR APPROVAL



In response to John's question, the following histogram shows the distribution of efficiencies for the zero bin.

The bins with zero efficiency are not included in this distribution, as they were assigned to the value -2. The remaining entries are randomly distributed efficiency values, and the peak around 50% is a consequence of the limited statistical sample.